

Finding New Cancer Uses for Existing Drugs

A heterogeneous graph neural network for drug repurposing on a biomedical knowledge graph – and an honest test of when the graph helps

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Research Proposal for Mentor Review | June 14, 2026

The problem. Bringing a new cancer drug to patients takes more than a decade and often fails. *Drug repurposing* – finding a new therapeutic use for an already-approved drug – is far faster and cheaper, and is especially valuable for cancers with few options. But the space of (drug, cancer) pairs is enormous and most candidates are dead ends. The key fact is that whether a drug might treat a cancer is rarely written in the drug’s own properties; it lives in a *web of relationships* – the drug’s protein targets, the pathways those proteins sit in, and the molecular processes the cancer disrupts. This project builds a model that reasons over that web to rank promising repurposing candidates, and – critically – tests honestly whether the graph structure is truly necessary or whether a simple feature-based model does just as well.

1 Background and Significance

Approved drugs have known safety profiles, so a credible repurposing hypothesis can reach patients years faster than a new molecule. The clinical precedent is strong: thalidomide for multiple myeloma, metformin in oncology trials, and kinase inhibitors crossing between cancer types. The challenge is *prioritization* – which of millions of (drug, cancer) pairs deserve costly experimental follow-up.

Whether a drug can treat a disease is fundamentally a *relational* property:

Why it matters	Consequence
Drug action is mediated by targets & pathways	The signal is in multi-hop links (drug → protein → pathway → disease), not a drug’s own descriptors.
Knowledge is fragmented	Integrating drugs, genes, pathways, phenotypes, and diseases into one graph exposes routes invisible to single-table analysis.
Most cancers have unexploited options	A ranked, evidence-backed shortlist focuses scarce experimental effort.

Table 1: Drug repurposing is a graph-reasoning problem: the evidence for a (drug, cancer) link is distributed across the biomedical network.

2 The Unmet Need

Gap 1 – Scale. The (drug $\times$ cancer) hypothesis space is too large to screen experimentally; computational prioritization is essential.

Gap 2 – Single-table models miss multi-hop biology. A model that sees only a drug’s and a disease’s own features cannot traverse the drug→target→pathway→disease chains that explain therapeutic action.

Gap 3 – Predictions are not actionable on their own. A bare ranked list without mechanistic rationale or literature support is hard for a researcher to trust or triage.

We exploit a simple observation: **a (drug, disease) link is a property of the two nodes’ positions in the biomedical network.** The information absent from a drug’s descriptors is present in the graph *around* it (Fig. 1). A model that propagates over the network can represent “this drug targets a protein in a pathway this cancer dysregulates”; a feature-only model must guess.

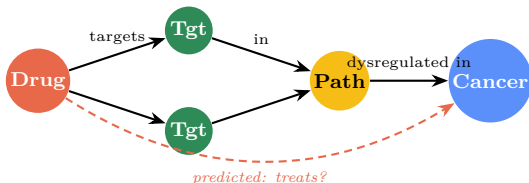


Figure 1: A repurposing hypothesis is a multi-hop path. The dashed link (does this drug treat this cancer?) is supported by a chain through shared targets and pathways – structure a feature-only model cannot see.

3 Research Aims

Aims

Aim 1. Build a heterogeneous graph neural network that predicts *drug* → *cancer* therapeutic links on a large biomedical knowledge graph and ranks novel repurposing candidates.

Aim 2. Test *honestly* whether graph structure is necessary – benchmark against a tuned tabular model on identical features, under leakage-safe inductive splits, with a topology-removal ablation.

Aim 3. Make predictions actionable: attach multi-hop mechanistic rationales and an automated, literature-grounded evidence report for the top candidates.

Guiding questions. (Q1) Can therapeutic links be predicted from a biomedical knowledge graph? (Q2) Does message passing beat a strong feature-only model, and *in which regime* (familiar vs. unseen diseases/ drugs)? (Q3) Are the top predictions mechanistically and bibliographically defensible?

4 Data

5 Proposed Methodology

We represent PrimeKG as a typed graph and pose repurposing as **link prediction** on the *drug* $\xrightarrow{\text{indication}}$ *disease* relation. Every node is given the *same* content feature – a sentence-transformer embedding of its name/description – so the comparison against non-graph models isolates the value

Resource	Content	Role
PrimeKG	~129k nodes (drugs, diseases, genes/proteins, pathways, phenotypes, ...), ~8M edges incl. indication / contraindication	Primary knowledge graph
Oncology subset	~3,000 neoplasm diseases flagged for focused evaluation	Cancer-specific targets
Europe PMC / PubMed	open literature search API	Evidence retrieval for the report layer

Table 2: All inputs are public. PrimeKG is purpose-built for precision-medicine AI, with rich drug–disease therapeutic edges other graphs lack.

of structure, not features.

5.1 Proposed model: a heterogeneous GNN

A heterogeneous GraphSAGE encoder produces a node embedding by pooling type-specific messages over L hops,

$$h_v^{(l)} = \sigma\left(\sum_{r \in \mathcal{R}} W_r^{(l)} \text{AGG}_{u \in \mathcal{N}_r(v)} h_u^{(l-1)}\right), \quad \hat{p}(d \rightarrow x) = \sigma(\phi(h_d, h_x)), \quad (1)$$

and a decoder ϕ scores a (drug d , disease x) pair from their embeddings. Because h_d and h_x are built from multi-hop neighbourhoods, the model can encode the drug→target→pathway→disease evidence of Fig. 1.

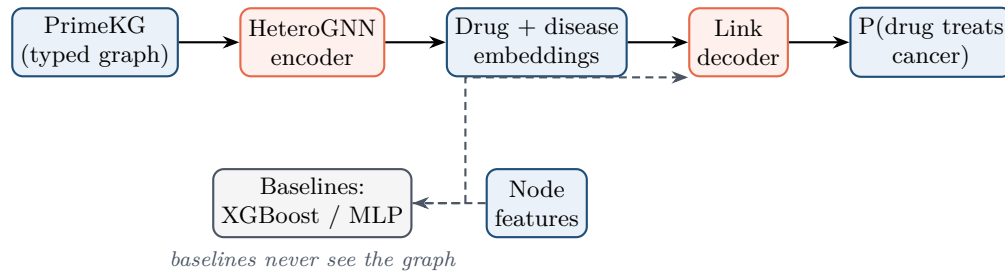


Figure 2: Pipeline. The GNN (orange) pools each node’s KG neighbourhood; the non-graph baselines (gray) score a pair from its two feature vectors alone.

5.2 Baselines and rigorous evaluation

Leakage-safe splits. (i) *Transductive* – random held-out links; all nodes seen. (ii) *Inductive cold-disease* – whole cancers held out (all their drug links removed from training), the disease-discovery regime. (iii) *Inductive cold-drug* – whole drugs held out. In every regime we remove each held-out therapeutic link from the message-passing graph in *both* directions (knowledge graphs often store **indication** bidirectionally, and leaving the reverse edge would let the GNN – but not the baselines – see the answer); negatives are sampled so they never collide with any known positive.

Model	Uses graph?	Purpose
HeteroGNN (proposed)	yes (multi-hop)	repurposing that uses network biology
XGBoost	no	strong tabular baseline on pair features
MLP	no	structure-blind neural control
DistMult (KGE)	memorized node embeddings	isolates memorization vs. generalization

Table 3: Every model receives the same node features; only the GNN sees the graph. The KGE has no representation for unseen nodes by design.

Hard negatives, not random. Random (drug, disease) negatives are trivially separable from true indications by node identity alone, inflating every model’s score. We instead use *class/area-matched* negatives (same drug class or same cancer family) and full per-disease candidate ranking, so the metrics reflect genuine prioritization difficulty.

Metrics. AUROC and AUPRC, plus *per-query* ranking metrics (Hits@ K and MRR computed by ranking each true link against its matched candidate negatives, not over a pooled set), as mean $\pm$ std over 5 seeds, with paired significance tests of the GNN vs. each baseline. The tabular baseline is hyperparameter-tuned so the comparison is against a genuinely strong model.

6 Rigor: Ablations, Interpretation, and an Evidence Report

6.1 Is the graph really doing the work? (falsification)

We run the identical GNN on the intact graph, a degree-preserving *shuffled* graph, and an *empty* graph. If the advantage is genuinely topological, destroying structure must collapse the GNN toward the feature-only baseline. A *relation ablation* (removing drug–protein, disease–protein, or pathway edges) then identifies which biological links carry the signal.

6.2 From predictions to an actionable shortlist

For the top candidates we (i) extract short multi-hop KG paths as mechanistic rationale – *down-weighting hub intermediates* (e.g. generic phenotypes like “fatigue”) so rationales are specific rather than popularity-driven, (ii) retrieve supporting/contradicting literature (Europe PMC), and (iii) use a large language model to draft a structured *evidence dossier* and an *LLM-as-judge* plausibility score that triages the list. To guard against the common failure where a model simply ranks globally popular drugs, we verify that the shortlist is *disease-specific* (top candidates differ across cancers) and report a quantitative hit-rate against a held-out time-split of known indications. The output is a ranked, citation-grounded set of cancer repurposing hypotheses.

7 Alignment with Strong-Research Criteria

8 Expected Outcomes

A proposal states targets, not measured results.

9 Eight-Week Timeline

10 Reproducibility and Ethics

Implementation uses open tools (PyTorch Geometric for the model; PrimeKG and Europe PMC for data). All code, configs, seeds, and a data manifest will be released; large artifacts are regenerated

Criterion	Commitment
Real, sizable public data	PrimeKG: $\sim 129k$ nodes, $\sim 8M$ edges.
Clear medical motivation	Faster, cheaper cancer therapy via repurposing.
Rigorous evaluation	Tuned XGBoost + MLP + KGE baselines; leak-free inductive splits; hard negatives; per-query ranking metrics; 5 seeds; significance tests.
Interpretability $\rightarrow$ discovery	Multi-hop rationales; literature-grounded evidence dossiers.
Ablations	Topology removal; relation ablation.
Translational impact	A ranked, vetted oncology repurposing shortlist.

Table 4: The project is structured around the elements of strong research.

Outcome	Target
Accurate repurposing	High AUROC/AUPRC on held-out therapeutic links.
Value of structure	$GNN \geq$ strong baselines; advantage clearest for unseen diseases/drugs, where memorization (KGE) collapses toward chance.
Causal check	Topology-removal ablation collapses the GNN to the feature baseline.
Actionable output	A ranked, mechanism- and literature-backed cancer repurposing shortlist.
Honesty clause	A strong tabular baseline is expected to be competitive; any narrowing of the gap is reported as-is, not overclaimed.

Table 5: Success is an accurate, rigorously-evaluated, interpretable repurposing method and a vetted candidate shortlist.

from public sources rather than committed. No human subjects are involved; all inputs are public and de-identified. Predictions are research-only and hypothesis-generating and do not constitute clinical guidance.

Selected References

1. Huang, Chandak, et al. *A foundation model for clinician-centered drug repurposing (TxGNN)*. Nature Medicine, 2024.
2. Chandak, Huang, Zitnik. *Building a knowledge graph to enable precision medicine (PrimeKG)*. Scientific Data, 2023.
3. Zitnik, Agrawal, Leskovec. *Modeling polypharmacy side effects with graph convolutional networks (Decagon)*. Bioinformatics, 2018.
4. Hamilton, Ying, Leskovec. *Inductive Representation Learning on Large Graphs (GraphSAGE)*. NeurIPS, 2017.
5. Yang, Trajanoski, et al. *Drug repurposing: progress, challenges and recommendations*. (review).

Weeks	Milestones
1–2	Build PrimeKG $\rightarrow$ PyG pipeline; shared node features; oncology subset.
3–4	Implement HeteroGNN + baselines (XGBoost, MLP, KGE); shared trainer.
5	Transductive + inductive (cold-disease/cold-drug) experiments, 5 seeds, significance tests.
6	Ablations: topology removal, relation ablation.
7	Interpretability + agentic evidence report; candidate shortlist.
8	Figures, tables, manuscript; reproducibility package.

Table 6: Schedule for the eight-week SRI cohort.